ENGINEERING REPORT · SPE1 · FULLY-IMPLICIT

SPE1 Fully-Implicit Review

Investigation `investigation_demo` · Report `report_1deb2…0ca0`

● DRAFT — NOT QUALIFIED

REPORT STEPS	SIMULATED WINDOW	NONLINEAR ITERATIONS	CONTENT SHA-256
2	86,400 – 259,200 s	7	3cc77fab… c2029

The investigation completed with **6** evidence-backed findings: **0** critical, **0** error, **2** warning, and **4** informational — drawn from reservoir-performance, economics, and history-match tool calls against one real, fully-implicit SPE1 simulation.

0	0	2	4
CRITICAL	ERROR	WARNING	INFO

✓ COMPOSITION AUDIT (REPORTCOMPOSITIONTOOL)

Report is not qualified; findings should be treated as draft pending human qualification. No missing recommendations, no missing limitations, no unverifiable evidence.

ECONOMICS

3 findings

WARNING economics.missing_price

Production of water has no assigned price; revenue excludes it.

claim calculated confidence 1.000 affected phase:water

WARNING economics.npv

Net present value over the saved observation window is negative (-5.00×10^{6}) under the supplied price and cost assumptions.

claim calculated confidence 1.000

INFO economics.summary

Net present value over the saved observation window is -5.00×10^{6} at a discount rate of 0.05; the undiscounted net cash flow is −334,741.

claim calculated confidence 1.000 affected field:economics

HISTORY MATCHING

1 finding

INFO history_matching.summary

History match compared 1 observed quantity against the stored result with a weighted-average normalized RMSE of 2.46×10^{-5}.

claim calculated confidence 1.000 affected field:history_match

RESERVOIR PERFORMANCE

2 findings

INFO performance.pressure

Mean reservoir pressure changed by 25,877.6 over the saved observation window.

claim calculated confidence 1.000 affected field:pressure

INFO performance.production

Integrated production over the saved observation window: oil produced 6,359.49, water produced 4.84×10^{-5}, gas produced 1.4385×10^{6}.

claim calculated confidence 1.000 affected field:production

EVIDENCE APPENDIX

3 references, 1 stored file

LABEL	URI	KIND	CHECKSUM
Reservoir performance for spe1_fim	result://spe1_fim/reservoir_performance	run	88e7424e…fac1f8
Economics for spe1_fim	result://spe1_fim/economics	run	88e7424e…fac1f8
History matching for spe1_fim	result://spe1_fim/history_matching	run	88e7424e…fac1f8

Generated by DeterministicOrchestrator → SingleCaseInvestigator → EngineeringReportBuilder from one real PyMRST SPE1 fully-implicit simulation run (2 report steps, 7 Newton iterations).

History-match observations are a perturbed copy of the real simulated field-average pressure, standing in for field data this demo does not have — not an actual historical measurement.